

BTCS501

V Semester Examination, December 2019

B. Tech./B.Tech.+MBA/B.Tech.+M.Tech. [IT]**Theory of Computation**

Choice Based Credit System (CBCS)

Time: 3 Hrs.

Maximum Marks : 60

Minimum Pass Marks: 24

Note: (1) All questions carry equal marks, out of which part 'A' and 'B' carry 3 marks each and part 'C' carries 6 marks.

(2) From each question, part 'A' and 'B' are compulsory and part 'C' has internal choice.

(3) Draw neat diagrams, wherever necessary.

(4) Assume suitable data, wherever necessary.

Q.1(A) Differentiate between transition diagram and transition matrix. 03

(B) Explain the basic structure of Finite state machine with diagram. 03

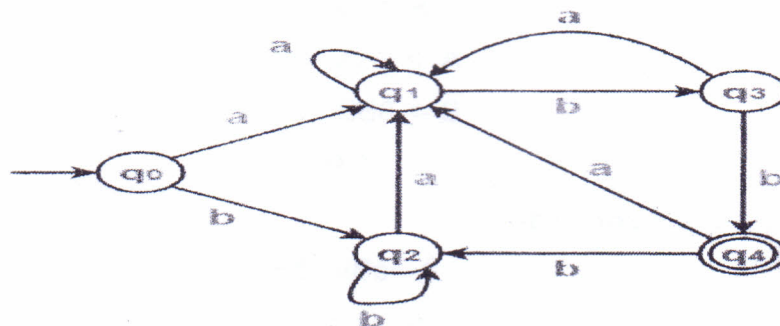
(C) Design DFA: $\Sigma = \{ 0, 1 \}$ 06

(a) $L = \{x \mid x \text{ start with } 110\}$

(b) $L = \{x \mid x \text{ end with } 1011\}$

OR

Minimize the following DFA.



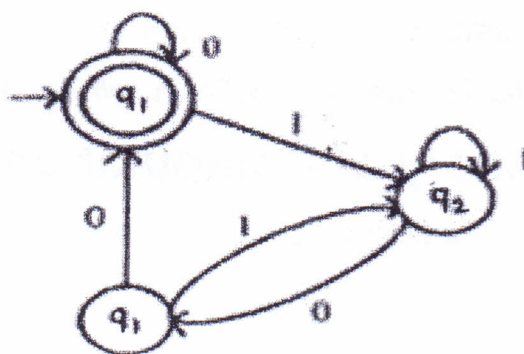
Q.2(A) Explain pumping lemma for regular languages. 03

(B) Describe the properties of regular expression. 03

Contd...

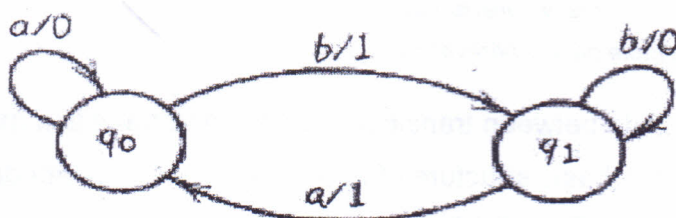
(C) Find out the regular expression from given DFA using Arden's theorem

06



OR

Convert the following Mealy machine into its equivalent Moore machine.



Q.3(A) Explain ambiguous grammar problems.

03

(B) Explain Greibach Normal form conversion steps.

03

(C) Convert following CFG to CNF

06

$S \rightarrow 1A \mid 0B$

$A \rightarrow 1AA \mid 0S \mid 0$

$B \rightarrow 0BB \mid 1S \mid 1$

OR

Simplify the given grammar

$A \rightarrow aBb \mid bBa$

$B \rightarrow bB \mid aB \mid \epsilon$

Q.4(A) Define PDA with the help of diagram.

03

(B) Explain how many ways PDA can accept input string.

03

(C) Explain the conversion steps of a final state PDA from the empty stack PDA.

06

Contd....

OR

Develop a PDA for $L = \{a^m b^{2m} \mid m > 0\}$.

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|--------|---|----|
| Q.5(A) | Define Turing Machine as Language acceptor. | 03 |
| (B) | Explain multi- tape and multi-head Turing machine. | 03 |
| (C) | Develop a Turing Machine for $L = \{wcw^R \mid w = (a+b)^*\}$. | 06 |

OR

Develop a Turing Machine for $L = \{1^m 2^m 3^m \mid m > 0\}$.

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